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Convolution analysis operator for multimodal image fusion

Chengfang Zhang^{a,*}

sichuan police college, 186 longtouguan Road, Luzhou, 646000, China

Abstract

Convolutional analysis operator learning, which takes advantage of the ability to extract and store overlapping blocks across training signals, has been the subject of much research in computer vision applications. The redundant filter learned by this method has the advantages of both constraining orthogonality and promoting diversity. This study, therefore, applies the convolution analysis operator to the field of image fusion and proposes a multimodal image-fusion method based on the convolution analysis operator. Experimental results show that this method performs better than the comparison methods as it not only retains the edges in the reconstructed image, but also considers the global structure of the image.

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Keywords: Multimodal image fusion; convolution analysis operator learning; edge preservation; convolutional sparse coding

1. Introduction

Multimodal images have many applications, including station security inspection and medical diagnosis. However, subject to different physical imaging mechanisms, the information from a unimodal image cannot adequately express all the information necessary for the scene. Multimodal image-fusion technology is particularly successful at obtaining more detailed information[1-5], and the obtained fusion images can facilitate target monitoring, recognition, and tracking in the later multimodal scene.

I. Y. Chun proposed a convolution analysis operator learning strategy[6-8]. The dictionary obtained by this strategy not only converges fast, but also ensures that the edges of the image reconstructed by the dictionary are well preserved. In this study, we leverage the superiority of the dictionary learned by the convolution analysis operator

* Corresponding author. Tel.: +86 0830-3166967.

E-mail address: chengfangzhang@scpolicec.edu.cn

and apply the dictionary to the image fusion of convolutional sparse coding. To ensure the performance of the proposed algorithm, we use three multimodal image-fusion algorithms as comparison methods, specifically, Z. Lei's algorithm for the fusion of infrared-visible images using non-sampled contourlet transform and area classification (NSCT-SSIM)[9]; the infrared-visible image fusion through gradient transfer and total change minimization algorithm (GTF)[10]; and B.K. Shreyamsha Kumar's cross bilateral filtering image-fusion method (CBF) [11] based on pixel importance. Our method and the three comparison methods are applied to two categories of multimodal images (infrared-visible images, and medical images). Experimental results show that our algorithm can better express image features.

2. Dictionary learning

In this study, I.Y. Chun's convolution analysis operator learning (CAOL) framework[7] with orthogonal constraints under the tight frame (TF) filtering condition (proposed by the same researcher) is used, and the USC-SIPI image data[1,2] set (50 frames of 512×512 standard images) applied to this dictionary learning framework (abbreviated as CAOL-TF) to obtain a compact and diverse dictionary filter (the dictionary filter size obtained in this study is $11 \times 11 \times 100$). Dictionary learning is expressed as follows:

$$\arg \min_D \min_{\{c_{l,k}\}} G(D, \{c_{l,k}\}) \quad s.t. \quad DD^H = \frac{1}{R} \cdot I. \quad (1)$$

$$G(D, \{c_{l,k}\}) := \sum_{l=1}^L \sum_{k=1}^K \frac{1}{2} \|d_k \otimes s_l - c_{l,k}\|_2^2 + \alpha \|c_{l,k}\|_2^2 \quad (2)$$

$$D := [d_1, \dots, d_k] \quad (3)$$

where $\otimes$ represents the convolution operator; $\{d_k \in \mathbb{C}^R: k = 1, \dots, K\}$ represents a group of convolution kernels; α is a threshold parameter that controls the sparsity of the feature $\{c_{l,k}\}$; $\{c_{l,k} \in \mathbb{C}^N: l = 1, \dots, L, k = 1, \dots, K\}$ represents a sparse coding group; and $\{s_l \in \mathbb{C}^N: l = 1, \dots, L\}$ is a training image set. The orthogonal condition $DD^H = \frac{1}{R} \cdot I$ satisfies $\sum_{k=1}^K \|d_k \otimes s\|_2^2 = \|x\|_2^2, \quad \forall s \in \mathbb{C}^N$.

3. The proposed fusion method

This study uses the dictionary learning algorithm proposed by I. Y. Chun to obtain a diversified convolution dictionary, $D_{CAOL-TF}$, and then applies the dictionary to convolutional sparse coding fusion. Fig.1 describe our fusion framework.

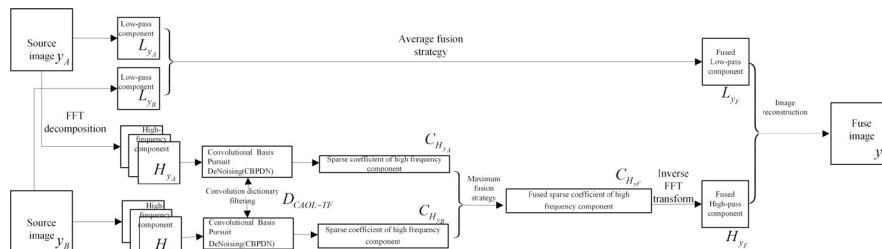


Fig. 1. Multi-modal image fusion framework.

4. Experiment and Analysis

To prove performance of algorithm, we compared the designed method with the other three fusion algorithms using two representative infrared-visible images, and medical images (as shown in Fig.2) as test images.

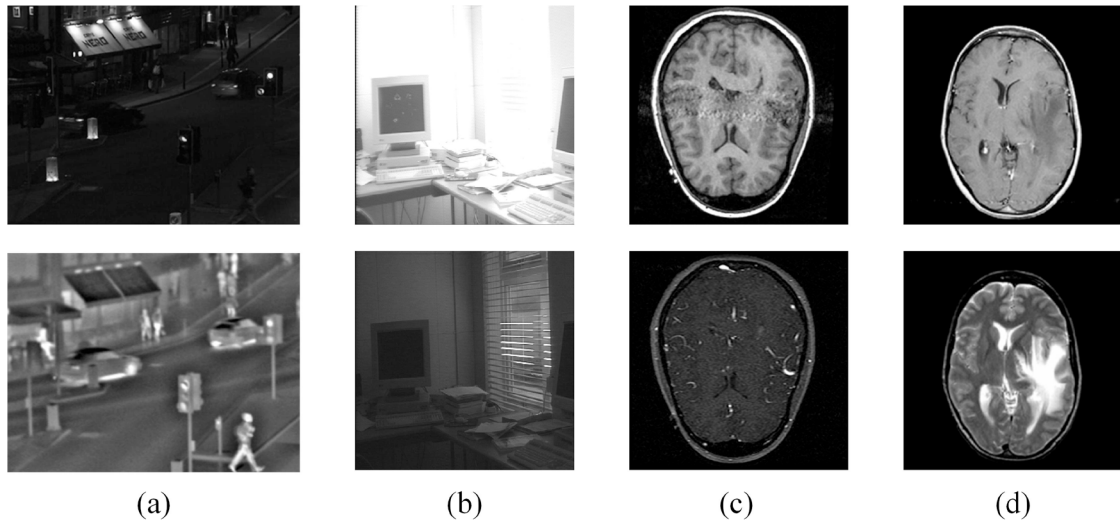


Fig. 2. Test source image.

4.1. Infrared and visible-image fusion results

Fig.3 and Fig.4 show the infrared-visible images fusion results: (a) indicates results obtained using the NSCT-SSIM fusion algorithm proposed by Zhang Lei; (b) indicates results obtained using the GTF fusion algorithm proposed by Ma Jiayi; (c) indicates results obtained using the cross bilateral filtering image-fusion algorithm (CBF) based on pixel importance proposed by BK Shreyamsha Kumar; and finally, (d) indicates results obtained using the fusion algorithm designed in this study. From the fusion results, it can be seen that the image-fusion quality from the application of NSCT-SSIM and CBF is the worst (see the pedestrians in Fig.3 (a), the road surface in Fig.3 (c), the computer in Fig.4 (a) and the wall in Fig.4 (c)); compared with the previous two methods, the GTF-based fusion results represent a significant improvement, but the infrared information is not better integrated (see Fig.3 (b) in which the cars and pedestrians can hardly be seen, and Fig.4 (b) in which the fused image demonstrates the phenomenon of exposure). Subjective observation shows our algorithm to be superior to the previous comparison methods.

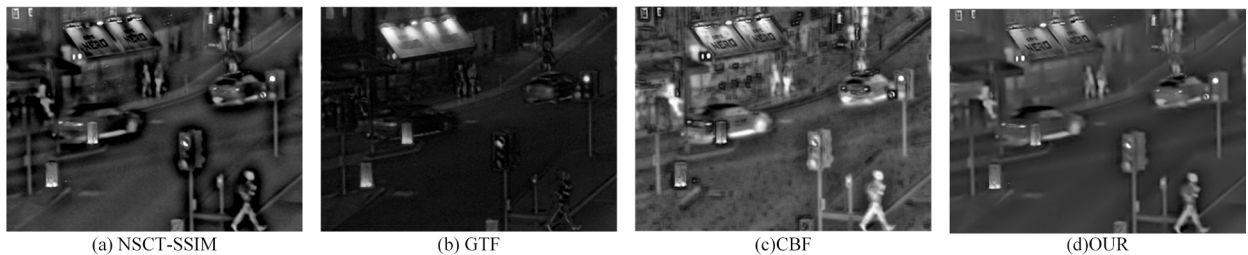


Fig. 3. Fused results of (a) group images applied to all methods.

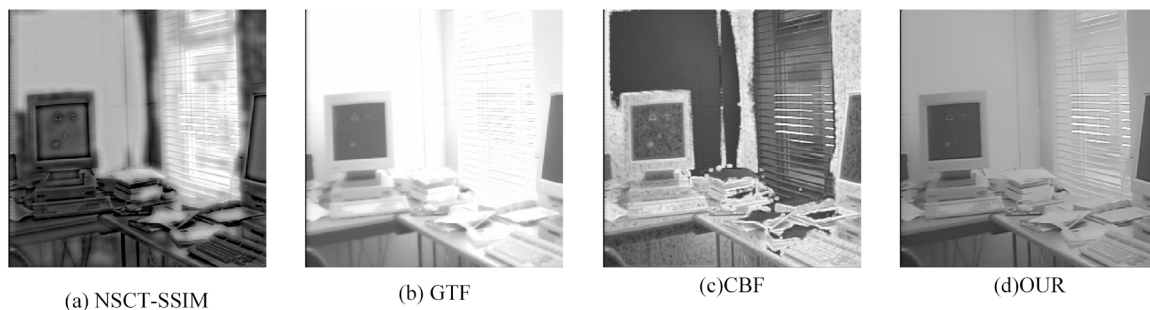


Fig. 4. Fused results of (b) group images applied to all methods.

4.2. Image-fusion results for medical images of the brain

Fig.5 and Fig.6 show the fusion results of two groups of medical images of the brain: (a) indicates results obtained using the NSCT-SSIM fusion algorithm proposed by Zhang Lei; (b) indicates results obtained using the GTF fusion algorithm proposed by Ma Jiayi; (c) indicates results obtained using the cross bilateral filtering image-fusion algorithm (CBF) based on pixel importance proposed by BK Shreyamsha Kumar; (d) indicates results obtained using the fusion algorithm proposed in this study. The methods based on NSCT-SSIM and CBF do not fuse soft tissue images well. The GTF-based fusion result is better than those obtained using NSCT-SSIM and CBF, but the overall brightness of the fusion image is low, which is not conducive to later diagnosis. Combining image results, it is clear that the proposed method can better integrate image information for images of soft tissue and bone.

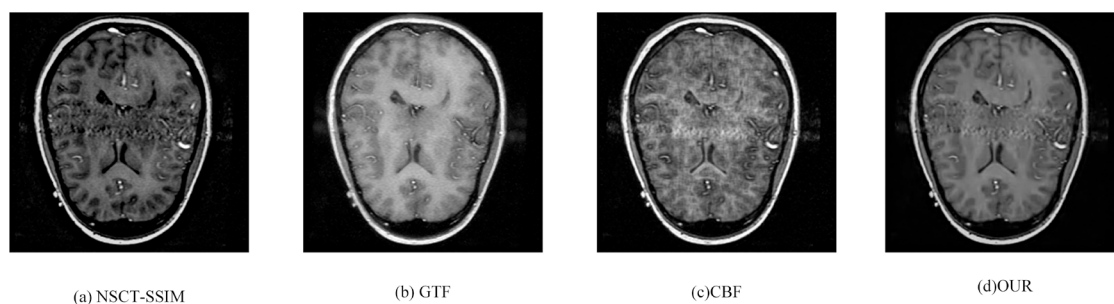


Fig. 5. Fused results of (c) group images applied to all methods.

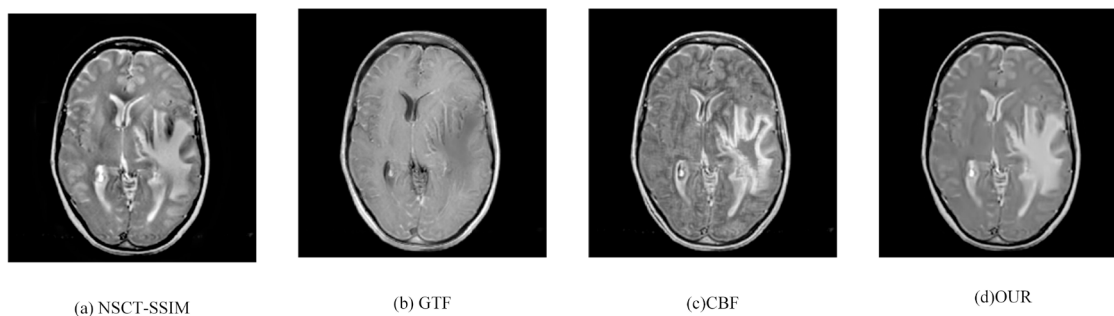


Fig. 6. Fused results of (d) group images applied to all methods.

4.3. Objective evaluation of image fusion

Tables 1-4 give the objective fusion evaluation for all comparison methods. Three Q-series objective evaluation indicators (Q^e [12,15], Q^{TE} [13,14], and Q^p [14,15]) are used to prove the superiority of the algorithm in this study, where the larger the value, the better the fusion effect (in this study, the highest value of the four indicators is highlighted in bold red font). It can be easily observed from the four tables that the evaluation index values obtained by our proposed method are the highest, much higher than those of the comparison methods; our method not only has the best edge fusion, but also integrates the source image features better.

Table 1. The objective evaluation of (a) group images.

	Q^e	Q^{TE}	Q^p
NSCT-SSIM ^[9]	0.3826	0.4268	0.3266
GTF ^[10]	0.2104	0.3674	0.1822
CBF ^[11]	0.2766	0.4195	0.2022
OUR	0.4219	0.4868	0.4478

Table 2. The objective evaluation of (b) group images.

	Q^e	Q^{TE}	Q^p
NSCT-SSIM ^[9]	0.4781	0.4759	0.3886
GTF ^[10]	0.3182	0.4521	0.2497
CBF ^[11]	0.4154	0.5193	0.2280
OUR	0.5443	0.5625	0.5388

Table 3. The objective evaluation of (c) group images.

	Q^e	Q^{TE}	Q^p
NSCT-SSIM ^[9]	0.3904	0.4947	0.3868
GTF ^[10]	0.3145	0.5414	0.2343
CBF ^[11]	0.3141	0.5571	0.2389
OUR	0.4444	0.5886	0.4909

Table 4. The objective evaluation of (d) group images.

	Q^e	Q^{TE}	Q^p
NSCT-SSIM ^[9]	0.3576	0.3597	0.3713
GTF ^[10]	0.2788	0.3586	0.1588
CBF ^[11]	0.3091	0.3666	0.2351
OUR	0.3967	0.3706	0.4376

5. Conclusion

Inspired by the convolution analysis operator learning (CAOL) framework to learn sparse regularization analysis from the perspective of convolution, we convolve the CAOL algorithm with orthogonal constraints of tight frame (TF) filter conditions from the convolution perspective. Dictionary filtering is applied to multimodal image fusion

based on convolution analysis operators. Compared with other fusion techniques, the convolutional sparse regularizer learned through CAOL-TF significantly improves the reconstruction quality of the fused image. Our fusion method is proven optimal in both subjective and objective evaluation.

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